

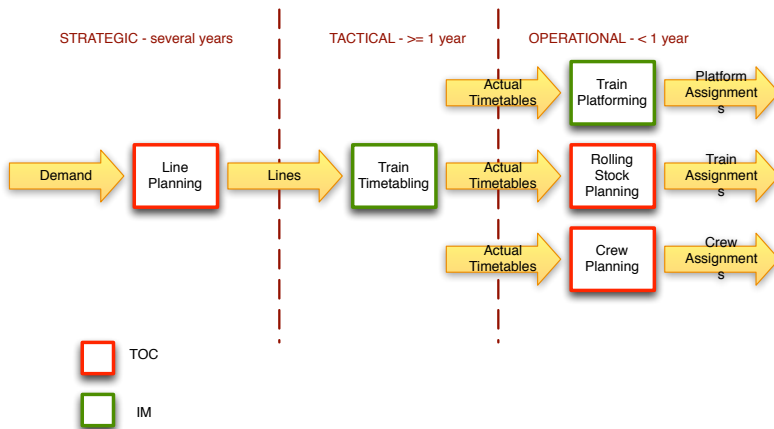
Demand Based Timetabling of Passenger Railway Service

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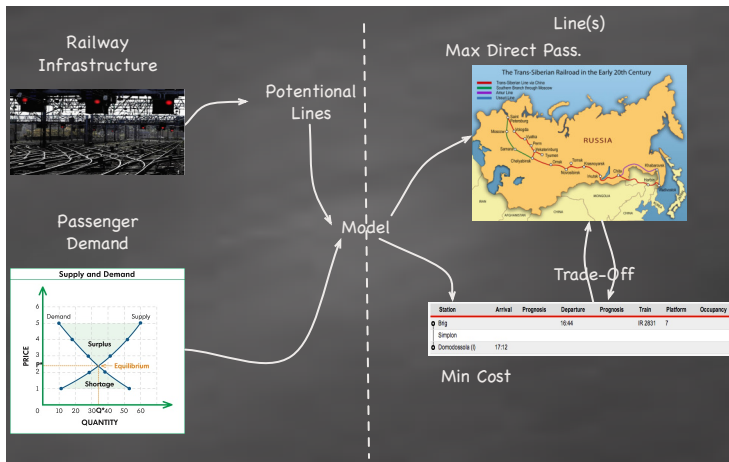
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for Research in Transportation,
Stockholm

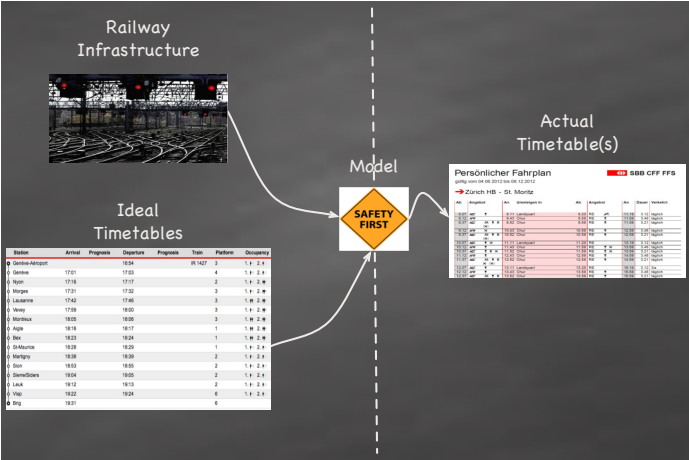
September 5, 2013

Railway Planning

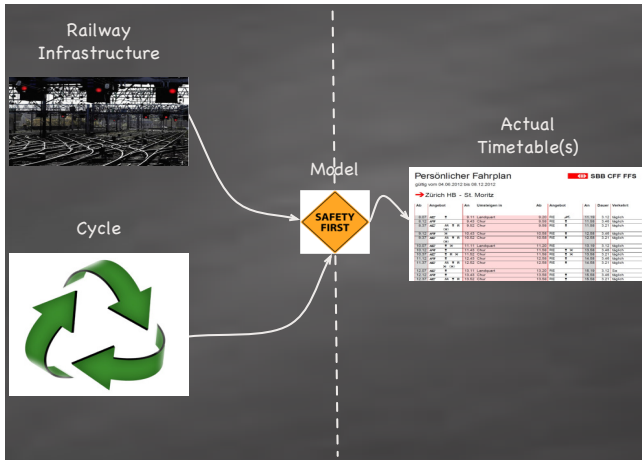


Line Planning Problem





Train Timetabling Problem – Cyclic



Arising Issues



Figure : Outside peak hour

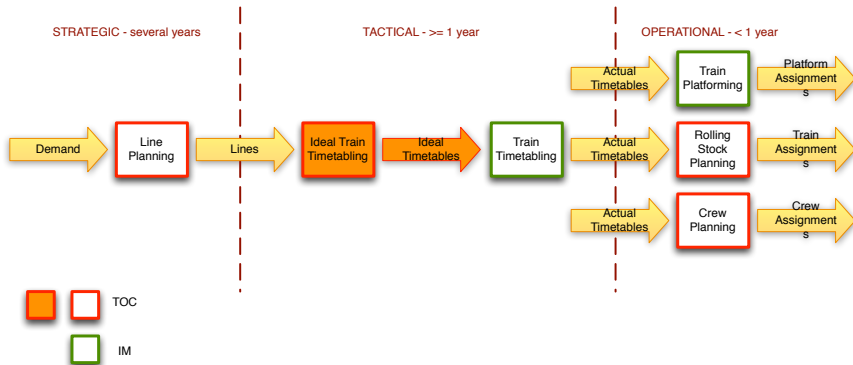


Figure : Inside peak hour



Figure : Train station in China

Railway Planning Improved



Agenda

- 1 Motivation
- 2 Ideal Train Timetabling Problem
- 3 Conclusions
- 4 Future Work

1 Motivation

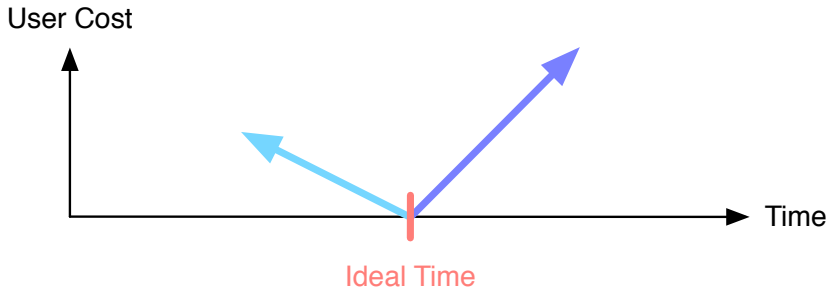
2 Ideal Train Timetabling Problem

- Assumptions
- Inputs
- Decision Variables
- Objective
- Constraints
- Cyclicity
- Connections

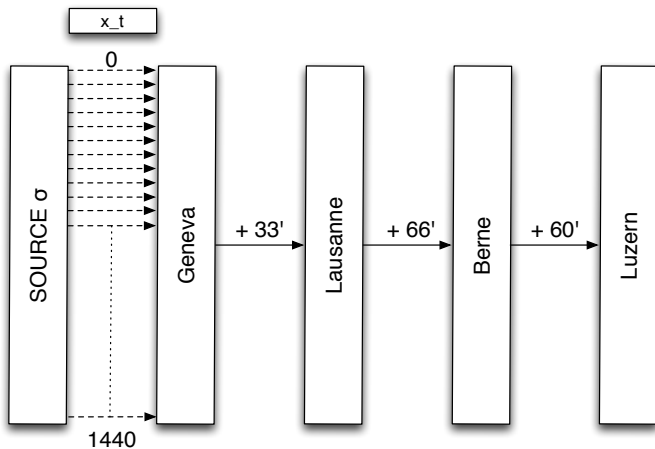
3 Conclusions

4 Future Work

Assumptions I



Assumptions II



Inputs

- $t \in T$ – set of time steps
- $l \in L$ – set of lines
- f – fraction by which it is better to be early
- d_t^l – demand captured along the line l , when scheduling a train at time t
- $d_t^{ll'}$ – connection demand captured along the line l and l' , when scheduling a train at time t on the line l
- n^l – number of trains available for line l
- $h_{ll'}$ – relative headway to reach a connection point of lines l and l' from the first station on line l and l'
- c^l – size of the cycle on line l
- s – preferred start of the planning horizon
- $M \in \mathbb{M}$ – set of sufficiently large numbers

Primary Decision(s)



$$x_t^I = \begin{cases} 1 & \text{if a train on line } I \\ & \text{is scheduled} \\ & \text{at time } t, \\ 0 & \text{otherwise.} \end{cases}$$

Secondary Decisions I



- $y_t^{lb} \in \mathbb{R}^+$ – cost of the passengers wanting to travel at time t on the line l , when taking a closest train at t or before
- $y_t^{la} \in \mathbb{R}^+$ – cost of the passengers wanting to travel at time t on the line l , when taking a closest train after t
- $y_t^l \in \mathbb{R}^+$ – cost of the passengers wanting to travel at time t on the line l

Secondary Decisions II



$$z_t^l = \begin{cases} 1 & \text{if passengers wanting} \\ & \text{to travel at time } t \\ & \text{on the line } l \text{ take the} \\ & \text{closest train} \\ & \text{after the time } t, \\ 0 & \text{otherwise.} \end{cases}$$

Objective

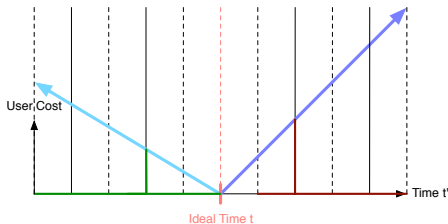
$$\min \sum_{l \in L} \sum_{t \in T} y_t^l \cdot d_t^l$$



Constraints I

$$y_t^{lb} \geq (t - t') / f \cdot \left(x_{t'}^l - \sum_{t''=t'+1}^t x_{t''}^l \right) \quad \forall l \in L, \forall t, \forall t' \in T : t \geq t',$$

$$y_t^{la} \geq (t' - t) \cdot \left(x_{t'}^l - \sum_{t''=t+1}^{t'-1} x_{t''}^l \right) \quad \forall l \in L, \forall t, \forall t' \in T : t < t',$$



----- Regular Time Step

— y_t^{lb}

— Departure

— y_t^{la}

Constraints II

$$y_t^{lb} \geq M_1 \cdot \left(1 - \sum_{t'=s}^t x_{t'}^l \right) \quad \forall l \in L, \forall t \in T,$$

$$y_t^{la} \geq M_1 \cdot \left(1 - \sum_{t'=t}^T x_{t'}^l \right) \quad \forall l \in L, \forall t \in T,$$

Constraints III

$$y_t^l \geq y_t^{lb} - z_t^l \cdot M_2 \quad \forall l \in L, \forall t \in T,$$

$$y_t^l \geq y_t^{la} - (1 - z_t^l) \cdot M_2 \quad \forall l \in L, \forall t \in T,$$

$$M_2 > M_1$$

Constraints IV

$$\sum_{t \in T} x_t^l \leq n^l \quad \forall l \in L,$$



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4 Future Work

Introducing Cyclicity

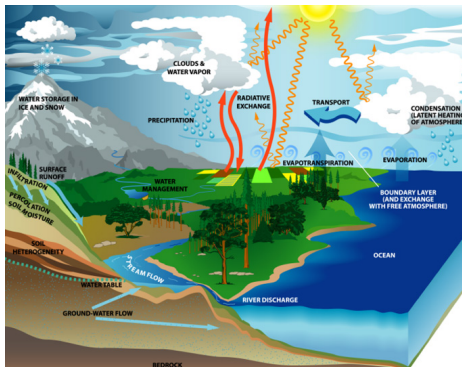
$$x_{t+c^l}^l = x_t^l$$

$$\min(t+c^l, T)$$

$$\sum_{t'=t+1}^{\min(t+c^l, T)} x_{t'}^l \leq (1 - x_t^l) \cdot M_3$$

$$\forall l \in L, \forall t \in T : t + c^l \leq T : t \geq s,$$

$$\forall l \in L, \forall t \in T : t \geq s,$$



Introducing Cyclicity

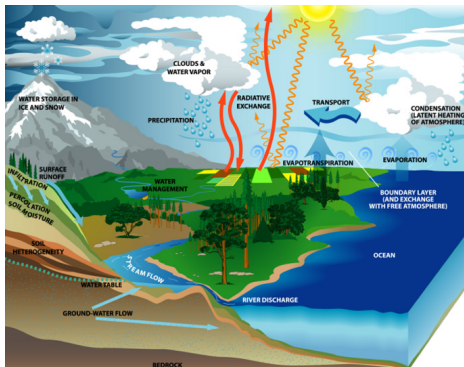
$$x_{t+c'}^l = x_t^l$$

$$\forall l \in L, \forall t \in T : t + c^l \leq T : t \geq s,$$

$$\min(t + c^l, T)$$

$$\sum_{t'=t+1}^{\min(t+c^l, T)} x_{t'}^l \leq (1 - x_t^l) \cdot M_3$$

$$\forall l \in L, \forall t \in T : t \geq s,$$



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Extra Decisions I



- $y_t^{l'b} \in \mathbb{R}^+$ – cost of the passengers wanting to travel at time t on the line l , when taking a closest train at t or before and connecting to line l'
- $y_t^{l'a} \in \mathbb{R}^+$ – cost of the passengers wanting to travel at time t on the line l , when taking a closest train after t and connecting to line l'
- $y_t^{ll'} \in \mathbb{R}^+$ – cost of the passengers wanting to travel at time t on the line l and connecting to line l'

Extra Decisions II



$$z_t^{ll'} = \begin{cases} 1 & \text{if passengers wanting} \\ & \text{to travel at time } t \\ & \text{on the line } l \text{ take the} \\ & \text{closest train} \\ & \text{after the time } t \text{ and} \\ & \text{connecting to line } l', \\ 0 & \text{otherwise.} \end{cases}$$

Objective

$$\min \sum_{l \in L} \sum_{t \in T} y_t^l \cdot d_t^l + \sum_{l \in L} \sum_{l' \in L} \sum_{t \in T} y_t^{ll'} \cdot d_t^{ll'}$$



Extra Constraints I

$$y_t^{ll'b} \geq (t - t') / f \cdot \left(x_{t'}^l - \sum_{t'''=t'+1}^t x_{t'''}^{l'''} \right) + \left(t'' - (t' + h_l'') \right).$$

$$\left(x_{t''}^{l''} - \sum_{t'''=t'+h_l''+1}^{t''-1} x_{t'''}^{l'''} \right) - M_4 \cdot \left(1 - x_{t'}^l + \sum_{t'''=t'+1}^t x_{t'''}^{l'''} \right)$$

$$\forall l, \forall l' \in L : l \neq l',$$

$$\forall t, \forall t', \forall t'' \in T : t \geq t' \text{ and } t' + h_l'' < t'',$$

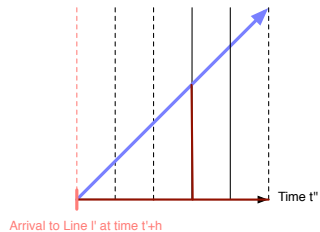
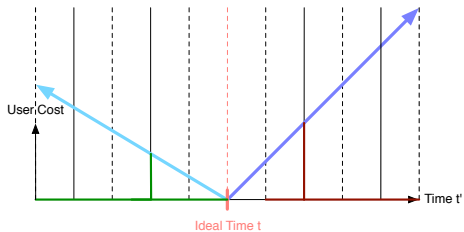
$$y_t^{ll'a} \geq (t' - t) \cdot \left(x_{t'}^l - \sum_{t'''=t+1}^{t'-1} x_{t'''}^{l'''} \right) + \left(t'' - (t' + h_l'') \right).$$

$$\left(x_{t''}^{l''} - \sum_{t'''=t'+h_l''+1}^{t''-1} x_{t'''}^{l'''} \right) - M_4 \cdot \left(1 - x_{t'}^l + \sum_{t'''=t+1}^{t'-1} x_{t'''}^{l'''} \right)$$

$$\forall l, \forall l' \in L : l \neq l',$$

$$\forall t, \forall t', \forall t'' \in T : t < t' \text{ and } t' + h_l'' < t'',$$

Extra Constraints II



----- Regular Time Step

----- $y'ilb$

----- Departure

----- $y'ila$

Extra Constraints III

$$\begin{aligned} y_t^{ll'} &\geq y_t^{ll'b} - z_t^{ll'} \cdot M_2 & \forall l, \forall l' \in L : l \neq l', \forall t \in T, \\ y_t^{ll'} &\geq y_t^{ll'a} - (1 - z_t^{ll'}) \cdot M_2 & \forall l, \forall l' \in L : l \neq l', \forall t \in T, \end{aligned}$$

Constraints to add

- Beginning and the end of horizon, when no connections are possible

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Conclusions

- New planning phase, based on the demand
- User cost rather than demand to capture (no need for discrete choice model)
- Can handle bot non- and cyclic timetables
- Connections are demand imposed

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Future Work

- Methodology design (cyclic is tighter than the non-)
- Actually solving the problem
- Analysis of the general results
- Analysis of the connections



Thank you for your attention.

References

The archived submission did not contain a separately labelled reference